

```
function updateNicknames(){
io.sockets.emit('usernames', nicknames);
}

socket.on('send message', function(data){
io.sockets.emit('new message', {msg: data, nick: socket.
nickname});
});

socket.on('disconnect', function(data){
if(!socket.nickname) return;
nicknames.splice(nicknames.indexOf(socket.nickname), 1);
updateNicknames();
});
});
```

The code is divided into three sections:

1. To start up the server
2. For user management
3. For sending and receiving messages

We should now discuss our client-side HTML file named *chat.html*:

```
<!DOCTYPE html>
<html>
<head>
<title>local chat</title>
<style type="text/css">
body{
font-weight: 400;
font-family: sans-serif;
}
h1{
font-weight: 300;
color: green;
text-align: center;
font-size: 24px;
}
#contentWrap{
display: none;
}
#chatWrap{
padding: 10px;
border: 2px solid black;
width: 300px;
box-sizing: border-box;
background: lightyellow;
float: left
}
#chat{
overflow-y: scroll;
height: 500px;
border: 1px solid green;
```



Figure 2: The result of the *test* application

```
width: 275px;
background: white;
padding: 0 10px;
box-sizing: border-box;
margin-bottom: 30px;
}
#chat ul{
list-style-type: none;
padding: 0;
display: inline-block;

}
#chat ul li{
padding: 4px;
box-sizing: border-box;
border: 1px solid blue;
background: lightblue;
margin: 1px 0;
font-size: 14px;
width: 250px;
}
#chat ul li b{
border-right: 2px solid #000;
margin: 0 5px 0 0;
}
#chatWrap form input{
height: 15px;
margin: 5px;
padding: 5px 5px;
width: 250px;
padding: 10px;
box-sizing: border-box;
}
#contentWrap #users{
margin-left: 320px;
width: 200px;
border: 1px solid black;
background: linear-gradient(120deg, #eee, #fff);
}

</style>
</head>
<body>
<h1>local chat</h1>

<div id="nickWrap">
<p>Enter a username:</p>
```